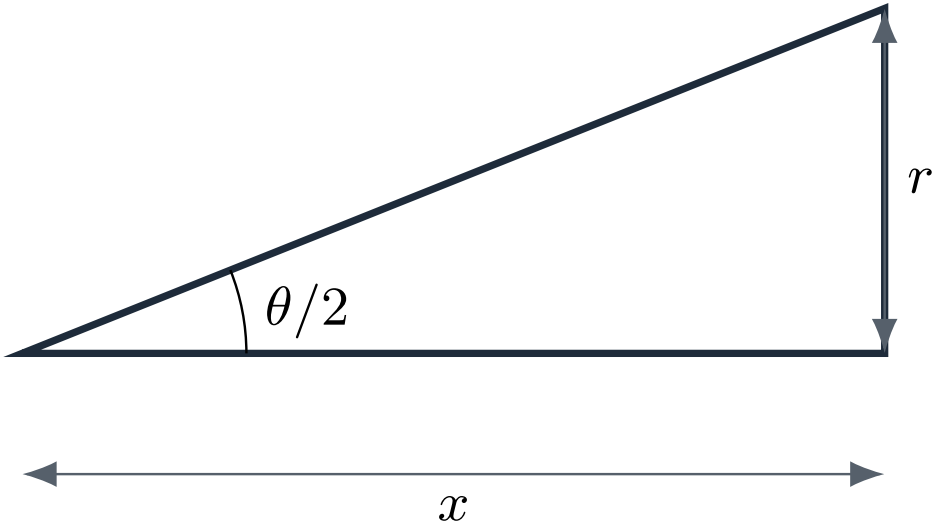

$$r = x\tan(\theta/2)$$

$$r_1 + r_2 = \text{const}$$

$$\omega_1 r_1 = \omega_2 r_2$$

长带近似与精确切线长度分开使用